

Q2

(a) First part of the FTC.

If f is a continuous integrable function on $\mathbb{R}$ and F is defined as

$$F(x) = \int_0^x f(t) dt$$

then $F(x)$ is differentiable and

$$F'(x) = f(x).$$

Second part of the FTC.

If f is continuous and integrable on $\mathbb{R}$ and F is an anti-derivative for f .

i.e. $F'(x) = f(x)$, then

$$\int_a^b f(x) dx = F(b) - F(a)$$

Applying the first part to the given problem. The function $f(t) = t^2 + \cos(t)$ is continuous and integrable on $\mathbb{R}$ so by the first part of the FTC. If

$$F(x) = \int_0^x t^2 + \cos(t) dt.$$

$$\text{then } F'(x) = f(x) = x^2 + \cos(x)$$

Let $N = \max(N_1, N_2)$. So how

whenever $\underline{n \geq N}$

$$|(\alpha a_n + \beta b_n) - (\alpha a + \beta b)|$$

" $< \varepsilon$ "

$$= |\alpha(a_n - a) + \beta(b_n - b)|$$

$$\leq |\alpha(a_n - a)| + |\beta(b_n - b)|$$

by the
triangle
inequality for
abs.-value

$$= |\alpha| |a_n - a| + |\beta| |b_n - b|$$

since abs. val. is
multiplicative, i.e. $|xy| = |x| |y|$

$$< |\alpha| \frac{\varepsilon}{2|\alpha|} + |\beta| \frac{\varepsilon}{2|\beta|}$$

$$= \varepsilon$$

This shows $\{\alpha a_n + \beta b_n\}_{n=1}^{\infty}$ satisfies
the definition for convergence to $\alpha a + \beta b$.

$= 6a^5$ by the linearity of limits of functions
as there is a factor of h^{-1} in all the
other terms

This holds for all $a \in \mathbb{R}$, $f'(a) = 6a^5$

So we can say the derivative is
the function defined by

$$f'(x) = 6x^5$$

b) $g(x) = \underbrace{x^4}_{\text{power}} \underbrace{\sin(2x)}_{\text{trig}}$

$$g'(x) = \frac{d}{dx}(x^4) \sin(2x) + x^4 \frac{d}{dx}(\sin(2x))$$

, by the product rule.

$$= 4x^3 \sin(2x) + x^4 \cos(2x) \cdot 2$$

, using standard derivative of
powers result and chain
rule on $\sin(2x)$.

For the second derivative we apply
these results again along with linearity to
get

$$g''(x) = \frac{d}{dx}(4x^3 \sin(2x)) + \frac{d}{dx}(2x^4 \cos(2x))$$

, by linearity.

$$= 12x^2 \sin(2x) + 4x^3 \cos(2x) 2$$

$$+ 8x^3 \cos(2x) + 2x^4 (-\sin(2x)) 2.$$

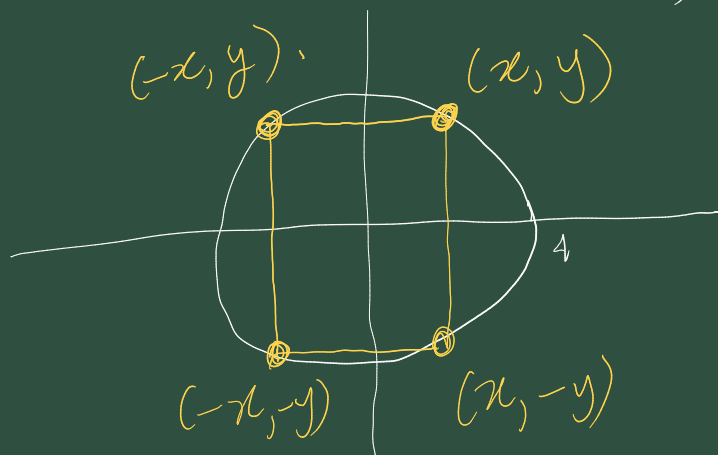
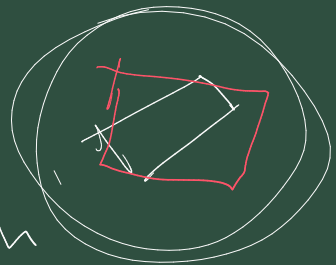
, using product and chain rules
like before.

$$= (-4x^4 + 12x^2) \sin(2x) + 16x^3 \cos(2x).$$

(c)

circle

Centre the unit circle on the origin
and rotate the rectangle so that it aligns
with the axes. A rectangle of maximal
area will touch the circle, at (x, y) .



The area A of this rectangle.

$$\text{is } A(x, y) = 2x \cdot 2y = 4xy.$$

